

Team 1

Iteration 0 Presentation

Team members

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Tech Stack

- Java
- Spring framework
- MariaDB / MySQL

- The project will be based on an existing project: <https://github.com/kim-0x/jpos> (created by our teammate Kimleng)
- The original project was a terminal based program, we will make some changes to it
 - Make a webui interface for it
 - Using Angular and Springboot

Existing Structure Strengths

- Role-Based Access Control: Covers basic permissions via users (Admin, Manager, Cashier).
- Historical Pricing: Uses a composite key (product_id, effective_at) in price_books to track historical selling prices.
- Line-Item Snapshots: Captures unit cost and unit price at time of purchase in sale_items, preserving historical transaction accuracy even when current product prices change.

Mock Payment Status

- Payment Processing & Checkout
- The `sale_transactions` table records the total purchase, but it lacks receipt status and payment method details.
- Payment Types: Create a `payments` table (or add columns to `sale_transactions`) to support split payments, cash, credit/debit card, or digital wallets.
- Transaction Status: Track payment states (completed, refunded, voided, pending).
- Cash Drawer Tracking: Record `amount_tendered` and `change_given` for cash transactions.

Inventory & Stock Management

- The inventory table tracks historical movements, but lacks real-time inventory logic.
- Current Stock Level: Add a `current_stock` column to `products` or build a view to sum inventory movements quickly without recalculating raw history on every page load.
- Movement Types: Add a `movement_type` column (restock, sale, damage_loss, return) to `inventory` instead of relying solely on positive/negative balances.
- Low Stock Alerts: Add `reorder_level` to `products` to alert staff when stock drops below a threshold.

Activities

- Team introduction, decided what project to do
- Task assignment
- Lab questions